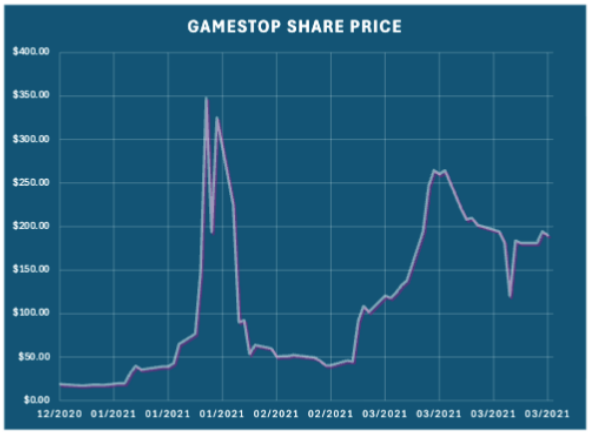
FNCE30007 DERIVATIVE SECURITIES

Week 4 Case Study - GameStop

1. Business Background

GameStop Corp. (NYSE: GME) is a US video game retailer that historically relied heavily on physical stores. Even before the COVID-19 pandemic, the company had been experiencing financial difficulties as consumers increasingly shifted from purchasing physical games to downloading games digitally. The pandemic placed further pressure on GameStop's store-based business model, while at the same time accelerating the broader shift toward online gaming and digital distribution.

By late 2020, these developments had created sharply different views among investors about GameStop's future. Some investors believed that the company's traditional business model was in structural decline and expected the share price to fall. Others believed that GameStop still had the potential to transform its business and benefit from a stronger e-commerce strategy.



January 2021 to March 2021

GameStop

2. The January 2021 Trading Frenzy

By the end of 2020, GameStop had become an unusually crowded trade. Many institutional investors believed that its traditional retail business was in decline and had taken large short positions in the stock. At one point in December 2020, reported short interest exceeded **138% of GameStop's shares outstanding**, making it one of the most heavily shorted stocks in the US.

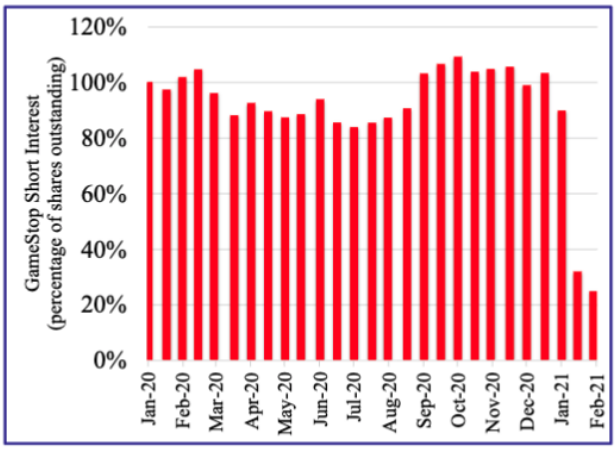


Figure 1: GameStop Short Interest

At the same time, a competing bullish view began to emerge. A prominent e-commerce entrepreneur and activist investor took a significant stake in GameStop, strengthening expectations that the company could successfully transform towards a more digital and e-commerce-focused business model. Retail investors also increasingly discussed GameStop on online communities and argued that the company could successfully transform its business. As attention grew, more investors began buying GME shares and call options, while also recognising how heavily the stock had been shorted.

This helped create a powerful feedback loop. Rising buying pressure pushed the share price higher, forcing some short sellers to buy shares to close their positions. That additional demand pushed the price up further, which attracted even more investors, media attention and market activity.

The speed of the move was extraordinary. On a pre-split basis, GME's share price and market capitalisation increased dramatically within just over two weeks:

Date	GME Closing Price	Market Capitalisation
11 Jan 2021	$19.94	$2.1 bn
13 Jan 2021	$31.40	$3.3 bn
22 Jan 2021	$65.01	$6.9 bn
26 Jan 2021	$147.98	$15.8 bn
27 Jan 2021	$347.51	$37.0 bn

In just over two weeks, GME's closing share price increased by more than 17 times.

The episode became known as the **GameStop short squeeze**.

3. The Short Squeeze & Put-call Parity

A short squeeze can occur when the price of a heavily shorted stock begins to rise, causing short sellers to incur losses and buy shares to close their positions. This additional buying demand can push the share price even higher, placing further pressure on remaining short sellers and creating a self-reinforcing cycle.

The GameStop short squeeze did more than create an extraordinary increase in the share price. It also created unusually difficult trading conditions. As demand for GME shares surged, short-selling became increasingly costly and constrained, making it more difficult for traders to implement arbitrage strategies.

This matters for option pricing. Put-call parity is based on a no-arbitrage relationship: when markets function normally, arbitrage trading should keep the prices of calls, puts and the underlying stock closely aligned. During extreme market conditions, however, trading frictions and short-selling constraints can make arbitrage difficult or costly, allowing larger pricing discrepancies to persist temporarily.

In the lecture, we will use actual GameStop option prices from January 2021 to examine what happened to the put-call parity during the short squeeze.

4. GameStop: Why Was the Apparent Arbitrage Opportunity Not Exploited?

Refer to the Excel file for all the price data.

On 27 January 2021, the observed prices were:

$S_0 = \$347.51, \quad K = \$200, \quad C = \$216, \quad P = \$88$

With $r \approx 0$, the American put no-arbitrage upper bound is

$P < C + K - S_0$

so

$P < 216 + 200 - 347.51 = \$68.49.$

However, the put was trading at **$88**, which is **$19.51 above the upper bound**.

So, was there an arbitrage opportunity?

In theory, yes. An arbitrageur could attempt to exploit the overpriced put by **selling the put, buying the call, buying a bond and short-selling the underlying stock** as part of the arbitrage strategy.

Under normal market conditions, such trading would put downward pressure on the put price and push the prices back toward the no-arbitrage relationship.

Why wasn't the opportunity immediately eliminated?

The problem was that **the arbitrage trade itself had become difficult and costly to execute**.

The strategy required the arbitrageur to **short GME shares**. During the short squeeze:

- **GME shares were difficult to borrow** because demand for borrowing shares was extremely high.
- **The cost of borrowing shares could be very high**, reducing or eliminating the apparent $19.51 profit.

- **GME's price was extremely volatile**, creating substantial losses and margin pressure on anyone holding a short position.
- **Trading and margin restrictions** could also make it difficult to establish or maintain the required positions.

Therefore, the $19.51 difference was an **apparent arbitrage profit**, not necessarily a **risk-free profit that could actually be locked in**.

The key lesson

The no-arbitrage argument assumes that traders can freely execute the required trades at reasonable costs. During the GameStop episode, this assumption was severely challenged.

Therefore, the GameStop case does not mean that put-call parity or no-arbitrage pricing was "wrong." It illustrates that theoretical no-arbitrage relationships rely on the ability of arbitrageurs to trade. When short-selling constraints, borrowing costs, liquidity problems and extreme volatility become severe, arbitrage may no longer be sufficiently attractive or feasible to eliminate the pricing discrepancy immediately.

5. Optional Viewing — *Dumb Money* (2023)

The story attracted worldwide attention and was later dramatised in the film ***Dumb Money***. Students who are interested in the broader story behind the case are strongly encouraged to watch the film.

6. Practice Questions

6.1 How Would You Trade GameStop?

POV: It is late 2020. You believe that GameStop can successfully improve its business by expanding its online operations and changing its business model. If the turnaround is successful, you expect investors to become more optimistic and GME's share price to rise substantially. However, given the uncertainty surrounding the company, you do not want to risk losing the full amount required to purchase GME shares.

Which of the following option positions (Long Call, Long Put, Short Put, Short Call) would best reflect your market view and risk preference? Explain your choice.

6.2 American Put–Call Parity in Current Market Conditions

For this question, use the current (August 2026) market data provided in the second worksheet (Tutorial - GME 2026) of *GameStop_Options_Case_FNCE30007.xlsx.*

(a) Using the data provided, calculate the lower and upper bounds for the American put option implied by the American put-call parity relationship.

(b) Compare the observed market put price with your calculated range. Does the observed put price fall within the no-arbitrage range?

(c) Compare your result with the January 2021 GameStop short-squeeze example discussed in the lecture. What do you notice, and what might explain the difference?